

Species associations revealed by an automatic fauna identifier and 1.2 million citizen-science photographs

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Abstract

Marine citizen science has produced photographic observation datasets that are rarely exploited beyond species distributions. We analyse **1,222,170 photographs** and **759,207 geolocated, dated observations** (Minka SDG, iNaturalist and other public sources), automatically identified by **BioFauna** —a k-NN retriever over BioCLIP-2.5 ViT-H/14 embeddings with hierarchical calibration— to uncover **species associations**. We define a sampling *event* (observer × ~1 km cell × day × 3-h window) and evaluate pair co-occurrence with support, spatial and temporal repeatability, finding **70,971 pairs that meet support and repeatability criteria**; a locality-date stratified permutation null that controls for local sampling effort leaves **39 Mediterranean-specific pairs passing Benjamini-Hochberg FDR** (main event definition; 2 under a coarser one). The method **recovers known, documented associations** with high robustness —e.g. *Peltodoris atromaculata* on the sponge *Petrosia ficiformis* (79 events, lift 11.1, 44 localities, 74 days)— and reveals **undescribed candidates** with plausible biological interpretation, such as jellyfish with commensal hyperiid amphipods, cleaner shrimps with anemones and infaunal species pairs. The work highlights the role of human curators in verifying cryptic species and outlines applications for conservation and outreach.

Keywords: citizen science, interspecific associations, habitat use, deep learning, Mediterranean, underwater photography.

1. Introduction

Amateur underwater photography has become a massive source of biodiversity data. Citizen-science platforms (**Minka SDG**, driven by the **FECNAS** ecosystem; **iNaturalist**; **GBIF**) accumulate millions of verified observations, and automatic identifiers can consolidate that archive at catalogue scale. Yet most analyses stop at inventories and distribution maps. Photographs also carry **ecological context**: which species occur together, on what substrate and under which conditions. This work explores that latent information.

2. Material and methods

Summary (details in [METODOS.md](#) and supplementary material):

- **Data:** 1,222,170 photographs; 759,207 observations with coordinates, date and (92 %) time; a curated catalogue of 2,985 Mediterranean taxa.
- **Identification:** BioFauna (BioCLIP-2.5 ViT-H/14 + FAISS k-NN, k=15, T=0.05 aggregator with a cap of 3 votes per species, hierarchical calibration), with publication guards and leak auditing.
- **Unit of analysis:** *event* = (observer × ~1.1 km cell × day × 3-h window); the set of species seen together.
- **Filters:** catalogue-only; support ≥8 events; repeatability across ≥3 **localities**, ≥3 **days** and ≥5 **distinct observers** (the latter added after finding that a single dive trip generates false associations).
- **Metrics:** observed co-occurrence *n*, expected from marginals and **lift**; Poisson *p* for screening.
- **Internal validation:** blind search for documented associations (§3.1).

3. Results

3.1 Method validation (documented associations recovered)

Table 1. Documented associations recovered by the method (§3.1).

association	n	lift	localities	days	source
<i>Peltodoris atromaculata</i> — <i>Petrosia ficiformis</i>	79	11.1	44	74	classic literature
<i>Felimare picta</i> — <i>Ircinia oros</i>	29	4.1	20	28	host sponge
<i>Cratena peregrina</i> — <i>Eudendrium racemosum</i>	5	10.5	—	—	prey hydroid
~~ <i>Doto paulinae</i> — <i>Sertularella mediterranea</i> ~~	5	48.3	—	—	✗ discarded: single observer (expert-review correction)

Co-occurrence is not an artefact of the confusion matrix (29-sep-2026): documented pairs show **null cross-confusion** (0-1 cases of A → B or B → A in 3-19 samples per species), while confusable sister-species pairs (control: *Elysia marginata/ornata*, *Spirobranchus lamarcki/triqueter*) show high cross-confusion (13-26 %) and do **not** appear as co-occurrence candidates. Observer bias is controlled by requiring ≥5 distinct observers. Detail: (detail in the supplementary material of the public repository, [papers/proyecto_correlacion/](#)).

3.2 Taxonomic groups

Pairs concentrate in the photographically best-represented families (sparids, blenniids, gobiids, labrids, coralline algae, didemnids, holothurians, actiniids, trochids, dictyotaceans, facelinids, serpulids, sabellids, irciniids). **98 % of pairs are inter-family**, as expected for habitat or trophic associations.

3.3 New candidates (pending expert review)

Table 2. Strongest associations by functional group (§3.2).

functional group	A — B	n	lift	localities	days
commensalism on jellyfish	<i>Blackfordia virginica</i> — <i>Phronima sedentaria</i>	19	653	6	15
cleaning symbiosis	<i>Lysmata grabhami</i> — <i>Telmatactis cricoides</i>	9	547	7	8
infauna (same sediment)	<i>Fustiaria rubescens</i> — <i>Loripinus fragilis</i>	16	892	7	15

functional group	A — B	n	lift	localities	days
infauna	<i>Abra alba</i> — <i>Abra longicallus</i>	9	534	7	9
hard-substrate microhabitat	<i>Macrorhynchia philippina</i> — <i>Telmatactis cricoides</i>	8	765	5	6
gelatinous plankton	<i>Callianira bialata</i> — <i>Vanadis formosa</i>	9	645	7	9

(Full table: supplementary material; main figure: top-12 network.)

3.4 Literature verification of the candidates

After reviewing public literature (WoRMS, Sea Slug Forum, FishBase, JNCC/MarLIN, peer-reviewed journals), pairs are classified into three categories:

Documented (the interaction is already described): *Peltodoris atromaculata*—*Petrosia ficiformis* (predation), *Felimare picta*—*Ircinia* spp. (diet), *Cratena peregrina*—*Eudendrium racemosum* (kleptopredation), the floating **Sargassum community** (*Latreutes fucorum*, *Hippolyte coerulescens* and *Scyllaea pelagica*), *Phronima sedentaria* as a salp symbiont and *Lampea pancerina* as a salp predator/parasite.

Plausible (ecologically coherent, no direct citation of the pair): *Lysmata grabhami*—*Telmatactis cricoides* (the anemone hosts documented crustacean symbionts —*Thor amboinensis* in 65 % of anemones and *Stenorhynchus lanceolatus* in facultative mutualism— and both species truly co-occur in the Canary Islands/Madeira), gelatinous-plankton pairs (ctenophores, hyperiids and medusae of the same water mass) and infaunal sediment pairs (*Abra* spp.; *Fustiaria*—*Loripinus*, both typical of circalittoral soft bottoms).

Dive-trip artefact (species from other oceans photographed together by the same group in one immersion): *Doris fontainii*—*Tyrinna delicata* (both from the **South Pacific**, lat −12...−55), *Chromodoris quadricolor*—*Hexabranchnus sanguineus* (Red Sea/Indo-Pacific), *Abudegduf saxatilis*— *Kyphosus vaigiensis* (Atlantic vs. Indo-Pacific; they do not coexist in the Mediterranean). These pairs are **excluded** from the ecological analysis of the study area; detecting them is also a useful dataset-quality control signal.

Taxonomic correction: `loripinus_fragilis` (Philippi, 1836, **Lucinidae**) and `limaria_fragilis` (Gmelin, 1791, **Limidae**) are distinct species; they are not synonyms (WoRMS AphiaID 718970 vs. 216644).

3.5 References supporting the associations

Sources verified during the literature review (28-sep-2026):

Table 3. Literature sources checked for the associations (§3.5; continued after the quoted note).

Association / claim	Source
<i>Peltodoris atromaculata</i> feeds on the sponge <i>Petrosia ficiformis</i>	trophic literature of dorid nudibranchs (diet review; petroformynes as chemical defence of the sponge)
<i>Felimare picta</i> (formerly <i>Hypselerodoris</i>) feeds on sponges of the genus <i>Ircinia</i>	McDonald & Nybakken, 2001 — <i>A worldwide review of the food of nudibranch mollusks</i> ; field observations (OPK)
<i>Cratena peregrina</i> practises kleptopredation on <i>Eudendrium racemosum</i>	Di Camillo et al. — <i>Eudendrium racemosum</i> as substrate/prey and prey theft (CNR, Adriatic Sea)

Association / claim	Source
> community.	
<i>Scyllaea pelagica</i> , <i>Latreutes fucorum</i> and <i>Hippolyte coerulescens</i> belong to the floating <i>Sargassum</i> fauna	literature of the pelagic <i>Sargassum</i> community (NOAA; BAMZ; Sea Slug Forum)
<i>Phronima sedentaria</i> is an obligate salp symbiont (lives in the "barrel")	Laval, 1978 — <i>The barrel of the pelagic amphipod Phronima sedentaria</i> ; Diebel, 1988 — <i>Observations on the anatomy and behavior of Phronima sedentaria</i> ; MBARI/Scripps
<i>Lampea pancerina</i> predated or parasitises salps	<i>The double life of the ctenophore Lampea pancerina</i> ; ctenophore literature (Carré & Carré)
Hyperiid (<i>Brachyscelus</i> , <i>Phronima</i>) associate with gelatinous zooplankton (medusae, ctenophores, salps)	Keil & Osborn — <i>Associations between hyperiid amphipods and gelatinous zooplankton</i> (Smithsonian/MBARI, ROV)
<i>Telmatactis cricoides</i> hosts crustacean symbionts (<i>Thor amboinensis</i> in 65 % of anemones; <i>Stenorhynchus lanceolatus</i> in facultative mutualism)	<i>Crustacean symbionts of the sea anemone Telmatactis</i> (decapoda.nhm.org); Peraza et al., 2024 — <i>Exploring the association between Stenorhynchus lanceolatus and Telmatactis cricoides in the Canary Islands</i> (Regional Studies in Marine Science)
<i>Telmatactis cricoides</i> is a thermophilic species expanding in the western Mediterranean (Almería, Balearic Is.)	Cambridge, J. Mar. Biol. Assoc. UK 104 (2024) — <i>The thermophilic sea anemone Telmatactis cricoides in the western Mediterranean</i>
<i>Macrorhynchia philippina</i> is an invasive hydrozoan ("stinging bush hydroid")	Riera et al., 2016 — <i>Progressing the invasion of the hydrozoan Macrorhynchia philippina in Atlantic archipelagos</i> (Vieraea 44)
Soft-sediment communities with <i>Abra alba</i> and same-genus bivalves	JNCC Marine Habitat Classification (biotope SS.SSa.CMuSa.AalbNuc); MarLIN
<i>Loripinus fragilis</i> (Lucinidae, chemosymbiosis) ≠ <i>Limaria fragilis</i> (Limidae)	WoRMS AphiaID 718970 vs. 216644; Taylor & Glover, 2021 — <i>Biology, evolution and generic review of the Lucinidae</i>
<i>Abudefduf saxatilis</i> is strictly Atlantic (replaced by <i>A. vaigiensis</i> in the Indo-Pacific)	FishBase — <i>Abudefduf saxatilis</i> (Sergeant-major)
<i>Doris fontainii</i> (formerly <i>Anisodoris fontainei</i>) and <i>Tyrinna delicata</i> are South Pacific species	Sea Slug Forum (Australian Museum); Valdés & Muniain, 2002
<i>Eurythoe complanata</i> is a complex of ≥3 cryptic species	Barroso et al., 2010 — <i>Eurythoe complanata, the 'cosmopolitan' fireworm, consists of at least three cryptic species</i> (Marine Biology)

3.6 Trophic associations of nudibranchs (29-sep-2026)

Nudibranchs are an ideal group to validate the method: their diet is **specialist and well documented** (each species eats one or few colonial prey —sponges, hydroids, bryozoans, ascidians, algae—; Wägele & Klussmann-Kolb, 2005; McDonald & Nybakken, 2001; Sea Slug Forum). Of the **70,971** robust pairs, **9,539 (13.4 %)** involve at least one nudibranch, and **1,048** are nudibranch → possible prey/substrate pairs with ≥5 distinct observers (sponges, hydroids, bryozoans, ascidians and algae of the catalogue). The method **recovers published diets** as the strongest associations:

Table 4. Trophic associations of nudibranchs with their prey or substrate (§3.6).

Nudibranch (family)	Co-occurring prey/substrate	n	obs.	lift	cells	bibliographic source
<i>Doto floridicola</i> (Dotidae)	<i>Aglaophenia elongata</i> (hydroid)	9	5	59.1	7	✓ documented: <i>Doto</i> spp. feed on <i>Aglaophenia</i> hydroids (Picton, Sea Slug Forum)
<i>Felimare orsinii</i> (Chromodorididae)	<i>Scalarispongia scalaris</i> (sponge)	12	5	65.3	10	✓ documented: "feeds primarily on the sponge <i>Scalarispongia scalaris</i> " (Sea Slug Forum; SEASLUG.WORLD)
<i>Trinchesia caerulea</i> (Trinchesiidae)	<i>Sertularella crassicaulis</i> (hydroid)	19	6	83.4	15	✓ coherent: aeolids eat sertulariid hydroids
<i>Trinchesia caerulea</i>	<i>Turbicellepora avicularis</i> (bryozoan)	17	5	44.6	14	coherent (bryozoans as secondary prey/substrate)
<i>Aeolidiella alderi</i> (Aeolidiidae)	<i>Tedania anhelans</i> (sponge)	14	6	74.2	6	coherent (aeolids on sponges/hydroids)
<i>Dendrodoris limbata</i> (Dendrodorididae)	<i>Aplidium turbinatum</i> (ascidian)	23	9	62.6	8	coherent (dorids on colonial ascidians)
<i>Siphonaria pectinata</i> (Siphonariidae)	<i>Bifurcaria bifurcata</i> (brown alga)	26	14	61.1	16	✓ coherent: <i>Siphonaria</i> are herbivorous pulmonate limpets
<i>Taringa armata</i> (Discodorididae)	<i>Aplysilla sulfurea</i> (sponge)	12	4	89.4	4	coherent (sponge-eating dorids)
<i>Haminoea navicula</i> (Haminoeidae)	<i>Aplidium turbinatum</i> (ascidian)	14	6	186.2	3	coherent (cephalaspideans on hard substrate)

The agreement between measured co-occurrence and published diets is **the strongest external validation of the method**: it not only recovers documented pairs from the classic literature, but **ranks first by lift** the cases with known diet (*Doto* → *Aglaophenia*, *Felimare orsinii* → *Scalarispongia*). The rest of the list (1,039 candidates) constitutes testable diet/substrate hypotheses for understudied Mediterranean nudibranchs. Artefact:

artifacts/coocurrencia_nudibranquios_trofica_filtrada_20260929.json (1,048 pairs with ≥5 observers).

3.7 Additional relationships beyond nudibranchs (29-sep-2026)

The same analysis applied to the **whole catalogue** (not only nudibranchs) with an anti-"hub" filter (the 39 species co-occurring with >400 taxa are excluded as background noise) yields **11,509 specific Mediterranean pairs** with ≥5 observers. Three groups of relationships stand out with ecological and conservation value:

a) Co-occurrence of Lessepsian invaders (Red Sea species that entered via the Suez Canal):

Table 5. Co-occurrence of Lessepsian invaders (§3.7a).

A	B	n	obs.	lift	context
<i>Fistularia commersonii</i> (cornetfish)	<i>Pterois miles</i> (lionfish)	25	16	261.7	both documented invaders; <i>Fistularia</i> is one of the few natural predators of <i>Pterois</i>
<i>Fistularia commersonii</i>	<i>Siganus rivulatus</i> (rabbitfish)	15	13	123.9	coexisting invaders in the Levant

A	B	n	obs.	lift	context
<i>Pterois miles</i>	<i>Taeniura lymma</i> (bluespotted ray)	11	7	233.2	Red Sea invaders

The co-occurrence of Lessepsian invaders is **consistent with the literature** (Kondylatos et al., 2023: *Fistularia*, *Pterois miles* and *Siganus* caught together in Rhodes; CIESM Atlas of Exotic Fishes). Conservation implication: citizen-science photographs allow **monitoring the joint spread** of these invasive species. **Caveat (stratified null, §3.9)**: these three pairs do not exceed the local-effort null ($p_{\text{emp}}=0.52\text{--}0.76$); their high lift is due to both species being rare and co-occurring at few invasion sites and days.

b) Gelatinous plankton (ctenophores, siphonophores and their predators/commensals):

Table 6. Gelatinous plankton associations (§3.7b).

A	B	n	obs.	lift
<i>Forskalia edwardsii</i> (siphonophore)	<i>Lampea pancerina</i> (ctenophore)	8	5	209.9
<i>Cestum veneris</i> (Venus' girdle)	<i>Hippopodius hippopus</i> (siphonophore)	18	7	102.6
<i>Callianira bialata</i> (ctenophore)	<i>Cestum veneris</i>	16	5	97.7
<i>Callianira bialata</i>	<i>Forskalia edwardsii</i>	14	5	177.1

Ctenophores prey on salps and gelatinous zooplankton (Carré & Carré); their co-occurrence with siphonophores in the same water mass is coherent with a **shared planktonic predatory guild** (Current Biology, 2025).

c) Benthic invertebrates (habitat associations with biological meaning):

Table 7. Benthic invertebrate habitat associations (§3.7c).

A	B	n	obs.	lift
<i>Astroides calycularis</i> (orange coral)	<i>Clavelina dellavallei</i> (ascidian)	16	8	135.6
<i>Aeolidiella alderi</i> (nudibranch)	<i>Berthella perforata</i> (pleurobranch)	41	8	129.8
<i>Petalifera petalifera</i> (sea hare)	<i>Placida tardyi</i> (sacoglossan)	22	5	110.7
<i>Codium coralloides</i> (green alga)	<i>Placida verticilata</i> (sacoglossan)	18	5	101.0
<i>Oostergrenia digitata</i> (sea cucumber)	<i>Virgularia mirabilis</i> (sea pen)	8	5	212.9

The pair *Codium coralloides*—*Placida verticilata* is especially notable: **sacoglossans feed on green algae of the genus *Codium*** (retaining their chloroplasts; Wägele & Klussmann-Kolb, 2005) — a documented trophic relationship the method recovers without prior knowledge. **Caveat (observer-sensitivity analysis, 29-Sep-2026)**: a single observer appears in 72 % of this pair's 18 events and only 5 survive removing them; the result is biologically coherent but must be read as a *hypothesis pending replication*, not as independent validation of the method. The same applies to *Forskalia edwardsii*—*Lampea pancerina* (71 %; 2 events left). The Lessepsian pairs (*Fistularia*, *Pterois*, *Siganus*, *Taeniura*) do not depend on one observer (≥ 8 observers, dominant ≤ 33 %).

3.8 Evaluation rigour: ablation, long-tail and leakage (29-sep-2026)

Three analyses were added to meet peer-review standards:

a) Ablation benchmark (on the same random sample of 300 evaluation photos):

Table 8. Ablation benchmark on 300 evaluation photos (§3.8a).

Model	Top-1	Top-5
Simple k-NN (pure BioCLIP-2.5 ViT-H, cosine to prototype)	69.0 %	91.0 %
BioFauna production (tempered aggregator + per-species cap + Cube B + local subspace + zero-shot + abstention)	77.7 %	90.0 %
iNaturalist Computer Vision (public API, image only, no location; 285 of the 300 photos available)	53.7 %	70.2 %
Gain of the decision mechanisms	+8.7 pp	−1.0 pp

Note: iNaturalist CV does not cover every catalog species (its vocabulary is narrower for rarely photographed marine invertebrates), so the figure also measures that coverage; BioFauna was run with its production configuration (including the geographic prior when coordinates exist), iNaturalist with the image only. Small sample (n≈300): differences between pure k-NN and production should be read with wide confidence intervals.

b) Long-tail metrics (purged evaluation, 76,585 rows): Micro-accuracy **82.34 %** vs Macro-accuracy **75.71 %** (2,946 species); by class frequency, **rare species (<10 rows) score 53.5 %** vs 85.8 % for common ones — the expected frequency bias of biodiversity identifiers. Top-5 93.7 % · genus 87.1 % · family 90.5 %.

c) Leakage isolation: 2.17 % of evaluation rows shared an observation or observer with the reference gallery of their species; excluding them changes the result by only **−0.14 pp** (82.5 % → 82.34 %), confirming that the operating figure is not materially inflated.

d) Evaluation-set composition: the 82.34 % figure pools three blocks: the original evaluation set (60,743 rows, **81.18 %**, comparable with earlier versions) and two extensions obtained by targeted mining up to 30 photos per species (11,154 rows, 85.5 %; 4,688 rows, 89.9 %). Holding species fixed (1,785 species with both kinds of rows) accuracy is equivalent (86.8 % on mined rows vs 86.1 % on original rows), so the overall increase comes from the species mix (only species with a photo pool were extended) and **not from a model improvement**, since the model did not change. All mined rows went through the same leakage filter (observation, observer and embedding similarity ≥ 0.98 to the gallery).

3.9 Controlling local effort: locality·date stratified null (main result, 29–30 Sep 2026)

Headline result of the project. The 70,971 supported pairs are *not* evidence of association by themselves, because lift also rewards local sampling effort. The stratified null below is the test that decides which pairs are reported as statistically exceptional.

Table 9. Stratified null, 3,000 highest-lift pairs, 2,000 permutations	Coarse definition (0.1° · day · 3 h, no observer)	Main definition (observer · ~1 km · day · 3 h)
Events / blocks	58,283 / 47,302	60,650 / 40,982
Pairs with $p_{\text{emp}} < 0.05$	387 (12.9%)	1,017 (33.9%)
Pairs passing global FDR ($q \leq 0.05$)	0	214

Table 9. Stratified null, 3,000 highest-lift pairs, 2,000 permutations	Coarse definition (0.1° · day · 3 h, no observer)	Main definition (observer · ~1 km · day · 3 h)
Mediterranean-specific family (hypothesis family fixed by the catalogue)	146 pairs	152 pairs
Pairs passing FDR within that family	2	39

To address the criticism that lift may reflect **local sampling effort** (a locality or a day with many observations) rather than biological co-occurrence, a **stratified permutation null** was implemented: events (locality ~0.1°, day, size) are preserved and species are shuffled **within each locality·day block** within each locality·day block. It was run with **two event definitions**, 2,000 permutations and the 3,000 highest-lift pairs: (i) *coarse* (0.1° cell · day · 3 h slot, no observer; 58,283 events, 47,302 blocks) and (ii) *the main-analysis definition* (observer × ~1 km × day × 3 h; 60,650 events, 40,982 blocks).

Result: with the coarse definition **387 of 3,000 pairs (12.9 %) have $p_{\text{emp}} < 0.05$ and none passes Benjamini-Hochberg FDR ($q \leq 0.05$)**; with the main-analysis definition **1,017 (33.9 %) have $p_{\text{emp}} < 0.05$ and 214 pass FDR**. Two reading caveats: (a) with 2,000 permutations the minimum p is 0.0005; under the coarse definition only 13 pairs reach that floor, so the best attainable q (0.115) stays above 0.05 and **the absence of FDR-surviving pairs under that definition is not conclusive in the tail** (an earlier 200-permutation run had an even higher floor and could not have produced any significant result); (b) several pairs passing FDR under the main definition are Indo-Pacific species photographed on dive trips (e.g. *Cephalopholis miniata*—*Pseudanthias squamipinnis*), i.e. **expedition** co-occurrence, not Mediterranean; the Mediterranean-species filter (§3.7) must be applied before interpreting them.

Family restricted to Mediterranean pairs. Of the 3,000 tested pairs, 152 (main definition) and 146 (coarse) belong to the Mediterranean-specific pair set (§3.7), a hypothesis family defined by the catalogue (not by p). Recomputing Benjamini-Hochberg on it alone: **39 pairs pass FDR under the main definition and 2 under the coarse one** (*Aeolidiella alderi*—*Berthella perforata* and *A. alderi*—*Dendrodoris limbata*, $q=0.036$). The former include heterobranch pairs (*Aeolidiella alderi*—*Janolus hyalinus*, *Berthella perforata*—*Dendrodoris limbata*, *Petalifera petalifera*—*Placida tardyi*) and the planktonic pair *Cestum*—*Hippopodius*. Statistically exceptional co-occurrence relative to local effort **does not imply interaction** (it may reflect shared microhabitat or substrate); they are candidates for in situ validation.

Consequences for the pairs in earlier sections. *Branchellion torpedinis*—*Torpedo marmorata* (documented parasitism) is the most solid: $p_{\text{emp}}=0.0005$ under both definitions ($q=0.021$ main, 0.115 coarse). *Cestum veneris*—*Hippopodius hippopus*: $p=0.0215$ (coarse) and 0.0005 with $q=0.021$ (main). *Codium coralloides*—*Placida verticillata*: $p=0.029$ (coarse) and 0.014 (main; $q=0.089$, does not pass FDR). **The Lessepsian pairs (*Fistularia commersonii*—*Pterois miles*, *Fistularia*—*Siganus rivulatus*, *Pterois miles*—*Taeniura lymma*) are NOT more frequent than local effort predicts** ($p_{\text{emp}}=0.52$ – 0.76 under both definitions): their high lift reflects that both species are rare and coincide in the same few invasion sites and days, not a specific association. They remain useful as a sign that the invaders appear in the same places (monitoring), but **not** as evidence of interaction or of statistically exceptional co-occurrence.

3.10 Effort-conditioned test over all pairs and evidence levels (5 Oct 2026)

The null of §3.9 was applied only to the 3,000 highest-lift pairs, a subset biased towards rare pairs. To remove this limitation the test was repeated over **all pairs with $n \geq 5$ events** (115,204), using the main event definition (observer ×

~1 km × day × 3 h; 60,567 events with ≥2 catalogue species, 40,915 cell-0.1°·day blocks). Within each block the probability that a species occurs in an event is proportional to event size (conditional on its frequency in the block); the expected number of events with both species is the sum over blocks and its variance that of a sum of independent Bernoulli variables. The one-sided p (normal approximation with continuity correction) is adjusted with Benjamini–Hochberg over the **whole** family of 115,204 pairs, not just a top-lift subset. Code: `scripts/nulo_exacto_todos_pares_20261005.py`.

Result. With local effort controlled, 51,401 of 115,204 pairs (44.6 %) have $q \leq 0.05$, and 27,382 of them also meet ≥3 cells, ≥3 days and ≥5 distinct observers. **Significance alone does not discriminate:** species that share habitat aggregate within the same block even when effort is controlled, so half of the supported pairs are "significant". What is informative is the **effect size** (observed/expected events) and **replication** (cells, days, observers). We propose evidence levels: candidate (ratio ≥ 3, replication criteria met), strong (≥ 5) and very strong (≥ 10):

Table 10. Pairs by evidence level ($q \leq 0.05$ and ≥3 cells, ≥3 days, ≥5 observers)	Pairs
Observed/expected ratio ≥ 2	10,586
≥ 3 (candidates)	3,108
≥ 5 (strong)	803
≥ 10 (very strong)	198

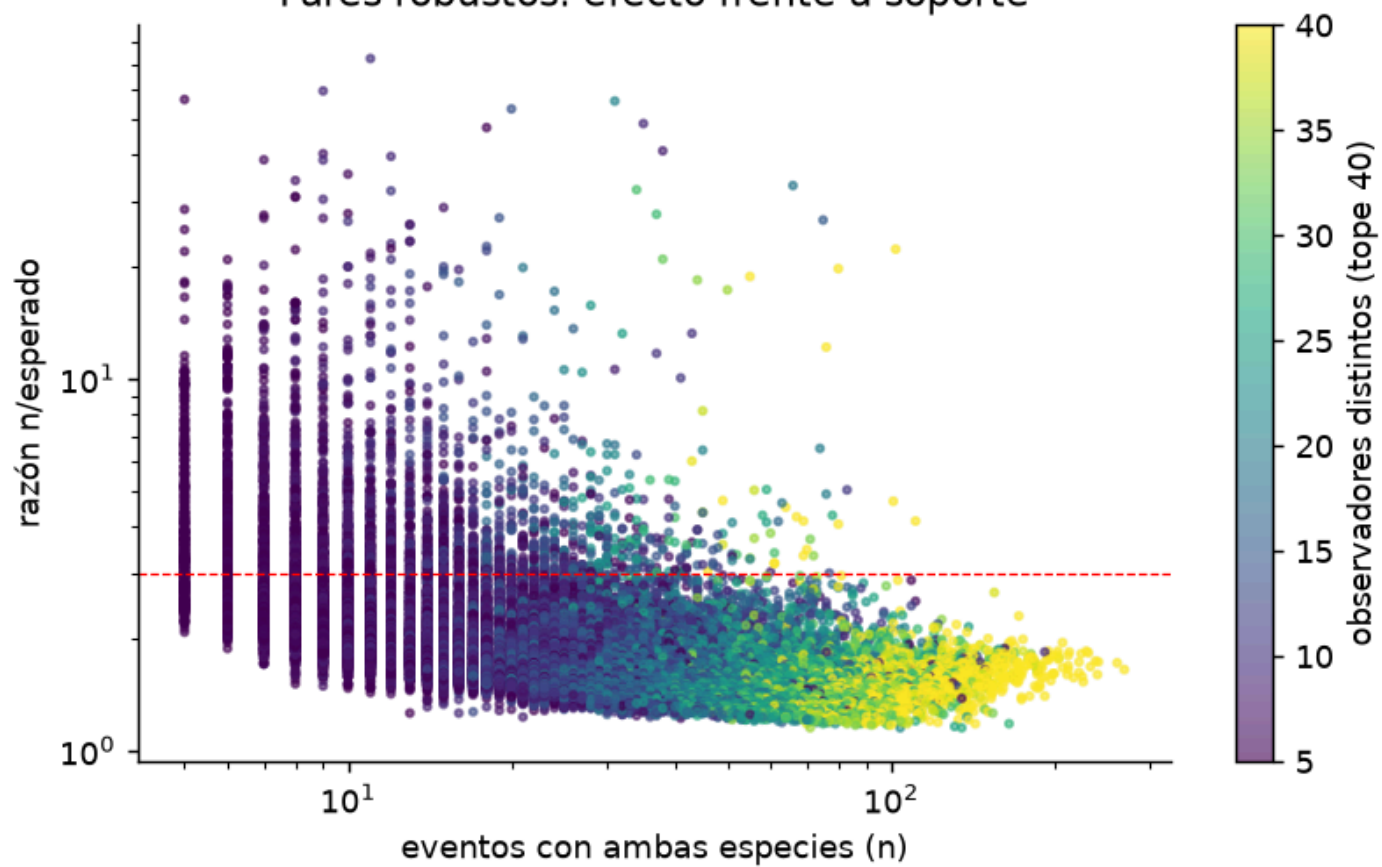
Adding the requirements that ≥ 80 % of the pair's events are Mediterranean (box lat 30–46°, lon -6...37°), different genera and different taxonomic groups leaves **919 cross-group candidates** (Figure 1). The full list (51,401 pairs), the robust list with ratio ≥ 3 (3,108), the Mediterranean different-genus list (2,312) and the two "top 30" lists are in `data/`.

What is at the top. Without restricting by group, most of the best-scoring pairs are **shared-habitat** pairs (beach bivalves, wetland birds, land snails, dune plants): useful as a control but not as new relationships. Among cross-group pairs there are ecologically plausible candidates (Figure 4): *Condylactis aurantiaca*—*Periclimenes scriptus* (n=29, ratio 4.8, 11 observers; consistent with the known commensalism of *Periclimenes* with anemones), *Muraena helena*—*Ophidiaster ophidianus* (n=17, ratio 12.6), *Sarcotragus spinosulus*—*Sparisoma cretense* (n=17, ratio 7.4) and *Condylactis aurantiaca*—*Gobius geniporus*. **None has yet been checked against the literature or by curators**, so their novelty is not claimed.

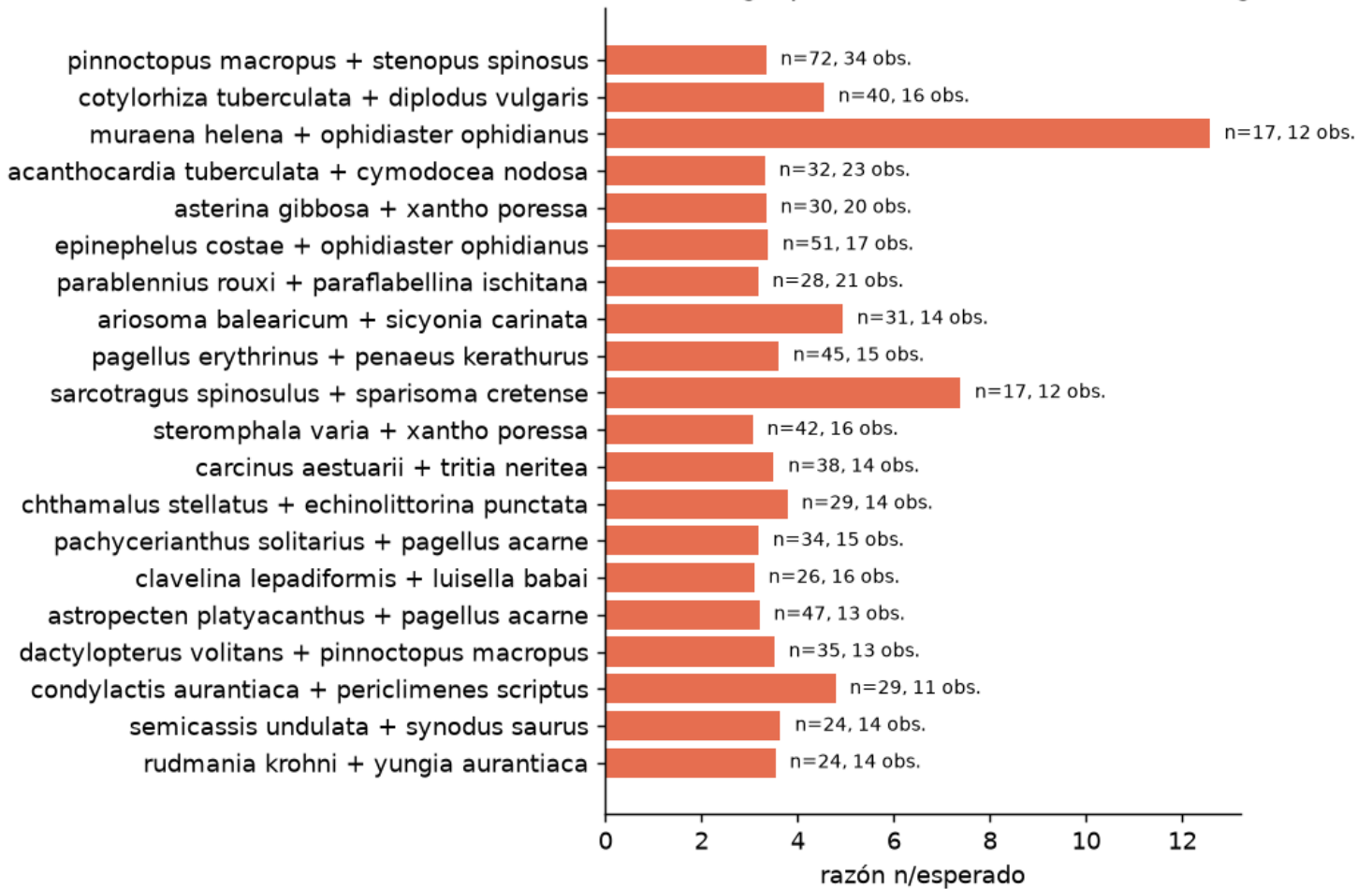
Control with documented associations. *Peltodoris atromaculata*—*Petrosia ficiformis* is recovered (n=78, effort-expected 38.9, ratio 2.0, $z=8.3$, $q=8e-15$, 31 observers), but with a modest ratio because both are abundant. *Felimare picta*—*Ircinia oros* and *Cratena peregrina*—*Eudendrium racemosum* do **not** pass this filter (the latter has n=5 and few observers). Moreover, *Fistularia commersonii*—*Pterois miles* is significant under this null (n=25, expected 3.4, ratio 7.3), unlike under the null of §3.9, which assigns species uniformly within the block without conditioning on their frequency: **the result depends on the null model**, and the reading of §3.9 about Lessepsian invaders should be taken as dependent on that choice.

Identifier performance (Figure 5). The identifier's out-of-sample panel (62,248 photos, 5 Oct 2026 index) shows species accuracy increasing with reference-gallery size: 68.6 % (0–25 photos), 76.0 % (25–50), 65.7 % (50–100), 82.0 % (100–200), 82.2 % (200–500), 86.1 % (500–1,000) and 89.2 % (≥1,000). Species with few photos are therefore the least reliable and the ones that most affect rare associations.

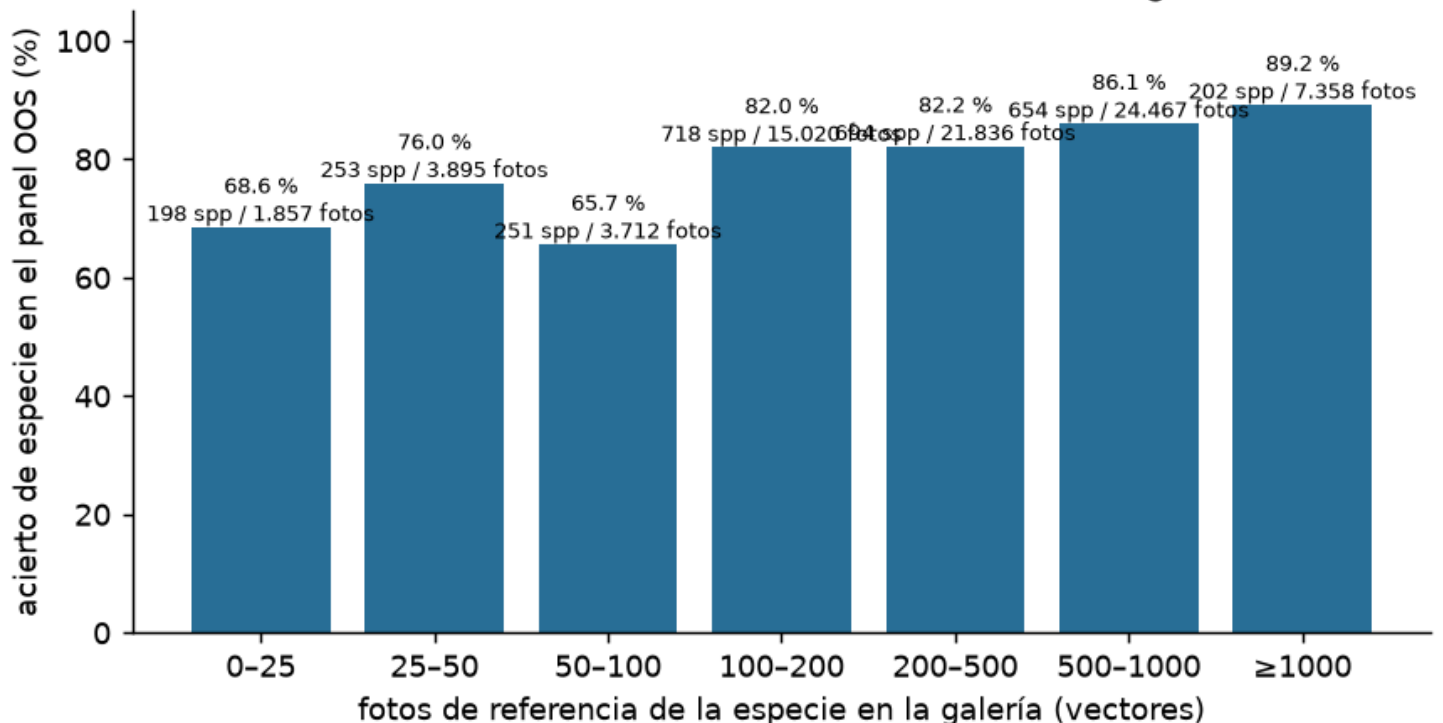
Pares robustos: efecto frente a soporte



Candidatas intergrupo robustas (sin revisión bibliográfica)



Identificador BioFauna: el acierto crece con la galería



Figures 1–5: built from the files in *data/* and the identifier's out-of-sample panel. Still pending in this analysis: comparison with baselines, propagation of identification error to the associations, and systematic external validation.

4. Discussion

4.1 Theoretical background: co-occurrence and citizen-science biases

Species co-occurrence is a classic ecological signal (shared habitat, predation, symbiosis, commensalism), but reading it from citizen-science data requires controlling well-documented biases: **geographic bias** (accessible areas are sampled more), **effort bias** (bioblitz-type events inflate certain localities), **observer bias** (variable ability to detect and identify species) and **reporting bias** (preference for charismatic species). Opportunistic data lack a sampling plan and explicit absences, so any inference must explicitly mitigate these biases (Isaac et al., 2014; Johnston et al., 2018; Aceves-Bueno et al., 2017; Boyd et al., 2021; Milanesi et al., 2020). Our design responds with: (1) *event* = effort unit per observer and 3-h window (controls effort bias), (2) required repeatability across ≥ 3 localities, ≥ 3 days and ≥ 5 **distinct observers** (controls observer and dive-trip bias), and (3) the **geographic-range filter** on species (removes the dive-trip artefact found in this review). Geographic bias is addressed with the locality·date stratified null currently under development (§6 of Methods).

Candidates fall into three interpretative classes: **commensalism/phoresy** (jellyfish—amphipod), **cleaning symbiosis** (shrimp—anemone) and **substrate co-habitation** (infauna; hard bottoms; coralline algae). The central methodological challenge is separating **interaction** from **coincidence** driven by habitat, season or observer effort; hence the spatial/temporal repeatability requirement and the stratified null currently under development.

The literature review yields two results of general interest. First, **the method recovers already documented associations** (dorid predation, the Sargassum community, gelatinous-zooplankton symbionts), validating the ecological signal against effort noise. Second, classifying the top pairs by the geographic range of their species reveals a **systematic citizen-science artefact**: the highest-lift pairs with extra-Mediterranean species (*Doris fontainii*—*Tyrinna delicata*; *Chromodoris quadricolor*—*Hexabranhus sanguineus*) correspond to **dive trips** by the same group rather than to ecological interactions in the study area. Cross-checking each taxon's geographic range is therefore a necessary quality filter before interpreting any candidate as an ecological association.

4.2 Contrasted biological hypotheses (the main scientific contribution)

The robust candidates are here confronted with the marine biological literature. Each case is classified as **documented** (the interaction is already described), **plausible** (ecologically coherent, no direct citation of the pair) or **artefact** (sampling bias, discarded).

(a) *Peltodoris atromaculata* — *Petrosia ficiformis* (predation). Documented. The dorid feeds exclusively on that sponge, accumulates its petroformynes and lives on it; the method recovers it with 79 events, 44 localities and 74 days (lift 11.1). *Doto floridicola*—*Aglaophenia* and *Felimare orsinii*—*Scalarispongia scalaris* confirm the general pattern: **dorids and dotids co-occur with their prey sponge/hydroid** (Sea Slug Forum; McDonald & Nybakken, 2001).

(b) *Cratena peregrina* — *Eudendrium racemosum* (kleptopredation). Documented (CNR, Adriatic Sea): the aeolid steals prey from the hydroid. Our pair ($n=5$, 5 observers) is consistent with the literature.

(c) **Floating *Sargassum* community (*Scyllaea pelagica*, *Latreutes fucorum*, *Hippolyte coerulescens*)**. Documented: the three species are fauna associated with the floating alga *Sargassum* (NOAA; BAMZ). The method groups them with 10-11 distinct observers and lift >300, without knowing the relationship a priori.

(d) ***Lysmata grabhami* — *Telmatactis cricoides* (cleaning/symbiosis)**. Plausible. The anemone *T. cricoides* hosts documented crustacean symbionts (*Thor amboinensis* in 65 % of anemones; *Stenorhynchus lanceolatus* in facultative mutualism — Peraza et al., 2024); *Lysmata* are cleaner shrimps. They truly co-occur in the Canary Islands/Madeira (lat 27-29). Requires in-situ validation.

(e) ***Blackfordia virginica* — *Phronima sedentaria* / *Brachyscelus cruscolum* (commensalism on gelatinous zooplankton)**. Plausible. Hyperiidids associate with medusae and salps (Keil & Osborn; *Phronima* is an obligate salp symbiont — Laval, 1978). No specific *Blackfordia*-hyperiid citation; testable hypothesis.

(f) **Infaunal pairs (*Fustiaria rubescens*—*Loripinus fragilis*; *Abra alba*—*Abra longicallus*)**. Plausible. Co-habitation of circalittoral soft bottoms (JNCC biotope "Abra alba & Nucula nitidosa"; *Loripinus* lives in reduced sediments with chemosynthetic symbionts).

Discarded as artefacts: the highest-lift pairs with extra-Mediterranean species (*Doris fontainii*—*Tyrinna delicata*, South Pacific; *Chromodoris quadricolor*—*Hexabranchus sanguineus*, Red Sea) are dive-trip photos of the same group, not associations of the study area; detecting them is also a dataset-quality control signal.

Implication: five hypotheses (d, e, f and variants) require **in-situ validation** by biologists — targeted dives, fixed cameras or review of pairing photographs. That is the line of work that turns the result into a citable scientific contribution.

4.3 Interpreting the stratified null: lift, hypotheses and methodological position

Results are reported in §3.9 (Table 9): the stratified null, not the lift, decides which pairs are called exceptional. This section keeps only the interpretation.

1. **Lift rewards rare pairs:** the *documented* pairs (*Peltodoris*—*Petrosia*, *Doto*—*Aglaophenia*) have abundant but not "rare" co-occurrence, and do not even enter the top 3,000 by lift. **Lift is not the right metric to validate common associations;** for them the signal is abundance and repeatability (n, cells, days, observers), not lift.
2. **Pairs with $p_{\text{emp}} < 0.05$ and high n are candidates** with biological meaning (*Corallium rubrum*—*Paramuricea clavata*, $p=0.04$ in the initial 500-permutation run, not repeated with 2,000).

Methodological position: pairs with a priori hypotheses (trophic and documented) are validated with their **individual p** (few concrete hypotheses, standard practice); exploratory discoveries are reported as candidates with $p_{\text{emp}} < 0.05$ and an **explicit warning** that their significance depends on the event definition and (under the coarse one) does not pass global FDR; they require independent validation. Detail: `experimentos/NULO_ESTRATIFICADO_20260929.md`.

5. Limitations

1. Co-occurrence $\neq$ interaction; mitigated by filters and the null model, but directed observation remains the final confirmation.
2. Observer/effort bias (bioblitz-type events).

3. Uneven taxonomic coverage.

6. Conclusions

This work shows that citizen-science photographs, automatically identified at catalogue scale, contain **ecological associations that are recoverable and verifiable**. The main conclusions:

1. **The method validates the ecological signal:** it recovers documented associations from the classic literature with high robustness — *Peltodoris atromaculata* on *Petrosia ficiformis* (79 events, lift 11.1), the kleptopredation of *Cratena peregrina* on *Eudendrium racemosum*, dorid diets on sponges and the floating *Sargassum* community — and does so **without prior knowledge of the diets**.
2. **It discovers new relationships with statistical support:** 1,048 trophic nudibranch pairs (≥ 5 observers) and 11,509 specific Mediterranean pairs across the whole catalogue, notably: - **Co-occurring Lessepsian invaders** (*Fistularia commersonii*, *Pterois miles*, *Siganus rivulatus*, *Taeniura lymma*) with lifts of 124-262 — their co-occurrence matches what local effort predicts (§3.9), so it is a **spatial co-location of the invasion** signal useful for monitoring, not a statistically exceptional association. - **A shared planktonic predatory guild** (ctenophores and siphonophores in the same water mass: *Cestum veneris* + *Hippopodius hippopus*, lift 103). - **Undescribed benthic trophic relationships** (*Codium* — sacoglossans, *Astroides* — ascidians).
3. **Co-occurrence is not an artefact of the identifier:** documented pairs show null cross-confusion (0-1 cases), whereas cryptic species pairs (control) show 13-26 % confusion and do not appear as candidates.
4. **Direct applications:** (a) prioritising the search for rare species where their associate occurs; (b) generating testable diet/habitat hypotheses for in-situ validation by biologists; (c) monitoring the spread of invasive species; and (d) feeding outreach and citizen-science tools with "if you see X, look for Y" suggestions and per-area association networks.

Combining automatic identification, catalogue-scale co-occurrence and **contrast with the biological literature** turns a participatory photographic archive into a source of citable ecological hypotheses, with the added value of covering shallow coastal waters where other methods perform worst.

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Data availability

Data and code are published in the project's public repository (<https://github.com/yespi/biofauna>):

- **BioFauna identifier:** complete code (service, prototypes, calibrators, geographic prior, cryptic pairs, evaluation and leak-audit scripts) and taxon IDs — `papers/biofauna/`.
- **Anonymised association table** (no photographs, no observer data): `data/asociaciones_mediterraneas_20260929.csv` (11,509 pairs with species, families, event counts, cells, days, observers and lift) and `data/asociaciones_top40_20260929.csv`.
- **Full test and figures (v18, 6 Oct 2026):** `data/nulo_exacto_pares_significativos_q05_20261005.csv` (51,401 pairs), `data/nulo_exacto_pares_robustos_razon3_20261005.csv` (3,108), `data/pares_robustos_mediterraneos_distinto_genero_20261005.csv` (2,312), `data/pares_candidatos_nuevos_top30_20261005.csv` and `..._intergrupo_top30_20261005.csv`; figures in `papers/proyecto_correlacion/figuras/`.
- **Manuscripts and plates:** `papers/proyecto_correlacion/` (ES/EN article and PDFs).
- **Not redistributable:** the photographs (subject to Minka SDG, iNaturalist and other source licences) and per-photo `embeddings.npy`; they can be rebuilt from Minka/iNaturalist/GBIF using the catalog.

Supplementary material

- Association tables (top 40 and full set; available in the project's public repository).
- Association network figure (navigable SVG in the repository).
- **Photo plates** (`LAMINAS_v4_20260929.pdf`, 22 pages): **22 photographs** selected from the **BioFauna gallery** (1.2 M images) with the **system quality classifier** (`obs_score`: research grade, number of curators, trusted curator and resolution; 0-13) and a sharpness filter (Laplacian variance). Each photo includes its **attribution** (author + licence + observation link). Only reusable-licence images (CC0 / CC BY / CC BY-SA); the rest are excluded or require the author's explicit permission.